

Computer Studies, Chapter 8 – Classes and Objects in Java

Exam Oriented MCQs, 8.1 Introduction

Q-1: Which of the following contains both data and program code?

- A. Object ~~B. Class~~ C. Attribute D. Function

Q-2: In Class, Data is referred to as...

- A. Program code B. Method C. Function ~~D. Attributes~~

Q-3: Function is referred to as...

- ~~A. Method~~ B. Class C. Object D. Attributes

8.2 Class in Java

~~Q-4: Which is a template for multiple objects with similar features?~~

- ~~A. Class~~ B. Attributes C. Method D. Object

Q-5: Which of the following embody all the features of a particular set of objects?

- A. Variable B. Object ~~C. Classes~~ D. Reference, variable

Q-6: Which keyword is used to define class in Java?

- A. Main ~~B. Class~~ C. Void D. Static

Q-7: Which of the following is a proper syntax of define class?

~~A. class <Class Name>~~

```
{  
<variable>  
<Method>  
}
```

C. Static main<Class name>

```
{  
<variable>  
<Method>  
}
```

~~A. Void <Class Name>~~

```
{  
<variable>  
<Method>  
}
```

D. Main <class name>

```
{  
<variable>  
<Method>  
}
```

Q-8: Java class is made up of how many components?

- A. One B. Two C. Three D. Four

Q-9: Which of the following define 'Attributes' in a class?

- A. Method B. Class C. Variables D. Behaviour

Q-10: Which of the following define 'behaviour' in a class?

- A. Methods B. Attributes C. Class D. Object

Q-11: What is used to access or modify attributes?

- A. Objects B. Behaviour C. Heap D. Methods

Q-12: In Object-Oriented Programming (OOP) languages, creating an object is also known as what?

- A. Object Instantiation B. Object creation C. Object making D. None of these

8.3 Creating Objects

Q-13: Which of the following are steps to create an object from a class?

- A. Declaration B. Instantiation C. Initialization D. All A, B & C

Q-14: In which step of creating an object, a variable name of type class is declared?

- A. Declaration B. Instantiation C. Initialization D. None of these

Q-15: What is the syntax to create class variable?

- A. <variable name> <class name> B. <object name> <class name>
C. <class name> <variable name> D. <object name> <variable name>

Q-16: In which step to create an object, keyword 'new' is used to create the object by allocating memory?

- A. Initialization B. Instantiation C. Declaration D. Delete object

Q-17: Which keyword is used to create an object by allocating memory?

- A. Create B. New C. Initialize D. Declare

Q-18: Which special method is called to initialize the newly created object?

- A. New B. Declare C. Class D. Constructor

Q-19: Which of the following can be specified by the class name?

- A. Type of variable name
- B. Type of formal parameter
- C. Return type of a function
- D. All A, B & C

Q-20: Which statement is used to declare an object of class 'Room'?

- A. object Room r1;
- B. Room r1;
- C. new Room r1;
- D. Room r1();

Q-21: Which of the following is the address of the memory location where the object is stored?

- A. 'new' keyword
- B. Heap
- C. Object
- D. Reference

Q-22: What is the name of a special memory portion where the objects live?

- A. Heap
- B. Class
- C. Object
- D. Hard disk

Q-23: Which method is executed when an object is created? OR

What is the name of the data structure, where all instances of class are allocated memory?

- A. Heap
- B. Object
- C. Constructor
- D. Void

Q-24: Why the special method 'constructor' is executed?

- A. To initialize variable
- B. To declare variable
- C. To clear memory
- D. To perform initial task

Q-25: When default constructor is called?

- A. With empty parentheses
- B. Without arguments
- C. Both A & B
- D. With curly brackets

Q-26: Which of the following can contain arguments that determine the initial values of variables?

- A. Curly brackets
- B. Parentheses
- C. New keyword
- D. Static keyword

Q-27: Which facility of Java looks for unused objects and reclaims the memory that those objects are using?

- A. Garbage collector
- B. Object
- C. Class
- D. Instance

Q-28: In Object Oriented Programming Languages, creating an object is called...

- A. Garbage collection
- B. Heap
- C. Reference variable
- D. Object Instantiation

Q-29: Which of the following is an abstract representation of an object?

- | A. Reference | B. Class | C. Attribute | D. Method |
|--------------------------|--------------------------|--------------------------|--------------------------|
| ☐ | ☐ | ☐ | ☐ |

Q-30: Which of the following is concrete representation?

- A. Instance B. Class C. Attribute D. Method

Q-31: Which variables are created at the time of creating an object and stay throughout the life of the object?

- A. Heap B. Garbage C. Method D. ~~Instance~~

Q-32: What is the name of method, which can be invoked using the objects to access or modify the instance variables?

- A. Instance object B. Referenced variables C. Instance methods D. Attributes

Q-33: What is used to define behaviour of an object?

- A. Class B. ~~Instance methods~~ C. Object D. Attributes

Q-34: In Java, which of the following returns a reference to an object that represents an instance of the class?

- A. Super B. New Keyword C. Extends D. Class Keyword

8.4 Accessing Instance variables and calling instance methods

Q-35: By using what instance variables and instance methods are accessed?

- A. Class B. Method C. ~~Objects~~ D. Operator

Q-36: Which operator is used to refer instance variables and instance methods?

- A. Is equal to (=) B. Plus (+) C. Dollar (\$) ~~D. Dot (.)~~

Q-37: What is the associativity of dot operator?

- A. Left to right B. Right to left C. Top to bottom D. Bottom to top

Q-38: What is the initial value of reference variable by default?

- A. Null B. 1 C. 0 D. None

8.5 Class Variables and Class Methods

Q-39: Which variables are called class variables?

- A. Method B. Static C. Void D. Null

Q-40: In which part value of class variables are stored?

- A. Class ~~itself~~
- B. Instance
- C. Memory allocated to object
- D. Anywhere

Q-41: Which variables can be accessed without creating instance of a class?

- A. Object
- B. ~~Static~~
- C. Method
- D. Void

Q-42: Which of the following is used to access class variables and class methods outside the class?

- A. <Method name>.<class name>
- B. <Class variable>.<class name>
- C. <method name>.<class variable>
- D. <class name>.<class variable/method name>

Q-43: Which of the following are global to the class itself and available to any other classes or objects?

- A. Methods
- B. Attributes
- C. ~~Class methods~~
- D. Class name

Q-44: The methods that operate on a particular object, or affect that object, should be defined as...

- A. Class method
- B. Static method
- C. ~~Instance method~~
- D. Void method

Q-45: Methods that provide some general utility but do not directly affect an instance of the class are better declared as...

- A. Instance method
- B. ~~Class method~~
- C. Void method
- D. Static method

8.6 Classification of variables declared in a class

Q-46: What are the variables defined inside method or blocks are called as?

- A. Class
- B. Method
- C. Instance
- D. ~~Local~~

Q-47: Which variables are created when the method or block is started and destroyed when the method or block has completed?

- A. Global
- B. ~~Local~~
- C. Class
- D. Instance

Q-48: Which variables are not initialized by default values?

- A. ~~Local~~
- B. Instance
- C. Class
- D. Global

Q-49: Which variables are defined within a class but outside any method?

- A. ~~Instance~~
- B. Local
- C. Global
- D. Static

Q-50: Which variables are allocated memory from heap area when an object is instantiated?

- A. Class ~~B. Instance~~ C. Non-static D. Static

Q-51: Which of the following variables are initialized by default values?

- A. Local B. Static ~~C. Instance~~ D. Non-static

Q-52: Which variables are defined with in a class?

- A. Local B. Instance C. Non-static ~~D. Class~~

Q-53: Which variables are defined outside any method, with the static keyword?

- ~~A. Class~~ B. Local C. Instance D. Method

Q-54: Which keyword is used to create class variables?

- A. Void B. Main ~~C. Static~~ D. Public

Q-55: Which variables are allocated memory only once per class and is shared by all its objects?

- A. Local ~~B. Class~~ C. Instance D. Private

8.7 Polymorphism

Q-56: What is the meaning of the word 'Polymorphism'?

- ~~A. Many Forms~~ B. Method's signature C. Method overwritten D. Many Reports

Q-57: What is the different methods having same name but a different signature called as?

- A. Many Reports B. Class Overloading
C. Class Overwritten ~~D. Method overloading~~

Q-58: Which of the following is a combination of the method name, the type of return value, a list of parameters?

- A. Class signature B. Object signature ~~C. Method signature~~ D. Variable signature

8.8 Constructors

Q-59: Which of the following is a special kind of method that is invoked when a new object is created?

- A. Class B. Object C. Method ~~D. Constructor~~

Q-60: Which of the following is mainly designed to perform initializing actions?

- ~~A. Constructor~~ B. Class C. Instance D. Method

Q-61: What are the default constructors called as?

- A. Empty constructor B. No-argument constructor C. Class name D. Instance variable

Q-62: Which constructor does not take any argument?

- A. Void B. Static C. Default D. Non-static

Q-63: Which of the following must have been the same name as class name?

- A. Object B. Constructor C. Void D. Instance

Q-64: Which of the following does not have return type?

- A. Instance B. Constructor C. Static variables D. Instance

Q-65: Which of the following is invoked implicitly only when an object is constructed using new operator?

- A. Constructor B. Local variables C. Methods D. Class variables

Q-66: How can we define a user-defined no-argument constructor?

- A. <classname>{}(); B. <objectname>{}(); C. <constructorname>{}() D. <classname>{}();

Q-67: Which of the following is not true for Java constructor?

- A. Constructor must have the same name as classname B. Default constructor does not take any argument
C. Constructor does not have return type D. Constructor can be invoked explicitly elsewhere in the program

8.9 Visibility Modifiers for Access Control

Q-68: From the following which type of visibility modifiers are used to protect a method or variables?

- A. Public B. Package C. Protected and Private D. All of them

Q-69: Which of the following is about controlling visibility?

- A. Package B. Access control C. Class D. Method

Q-70: Which of the following is about controlling visibility?

- A. Private Variables B. Methods C. Access modifiers D. Packages

Q-71: What is used to organize classes?

- A. Package B. Access control C. Constructor D. Access modifier

Q-72: How many levels of visibility are used to provide necessary protection?

- A. One B. Two C. Three ~~D. Four~~

Q-73: Which type of access should be defined, when we want to make method or variable visible to all the classes outside?

- A. Package ~~B. Public~~ C. Private D. Accessor

Q-74: In Java, which of the following is not a part of four p's of protection?

- ~~A. Primary~~ B. Private C. Public D. Package

Q-75: Which access is the widest possible access?

- ~~A. Public~~ B. Private C. Package D. Limited

Q-76: What is the syntax of Package statement?

- A. class<packagename>; B. object <packagename>;
C. package <packagename>; D. method <packagename>;

Q-77: Which access modifier is used to provide public access?

- A. Private B. Protected ~~C. Public~~ D. Package

Q-78: Which kind of variables and methods are visible anywhere? OR

Which access modifier in Java provides visibility to classes defined in another package also?

- A. Private B. Package C. Public D. Getter

Q-79: Which of the following is default level protection?

- A. Public ~~B. Package~~ C. Private D. Protected

Q-80: Which variable's scope is narrower than public variables?

- ~~A. Package~~ B. Public C. Private D. Protected

Q-81: Which level of protection is used to allow the access only to subclass?

- A. Private B. Package C. Public ~~D. Protected~~

Q-82: Which level of protection is used to share with the methods declared as "friend"?

- A. Package B. Public C. Private ~~D. Protected~~

Q-83: Use of which protection of Java will become more relevant when we use inheritance concept of object-oriented programming?

- A. Public B. Private ~~C. Protected~~ D. Package

Q-84: Which of the following protection's visibility is wider than full privacy provided by "private" level?

- A. Public B. Package ~~C. Protected~~ D. Private

Q-85: Which protection can be used to achieve highest level of protection?

- A. Package ~~B. Private~~ C. Protected D. Public

Q-86: ~~Which~~ of the following provide narrowest visibility?

- A. ~~Private~~ B. Public C. Protected D. Package

Q-87: ~~Which~~ methods and variables are directly accessible only by the methods defined within a class?

- A. ~~Private~~ B. Public C. Getter D. Setter

Q-88: Which variables can not be seen by any other class?

- A. Public B. Protected ~~C. Private~~ D. Both A & B

Q-89: Which of the following is extremely restrictive level of protection?

- A. Public B. Protected C. Package ~~D. Private~~

Q-90: Which is commonly used level of protection?

- A. Public ~~B. Private~~ C. Protected D. Package

Q-91: Which facilities are provided by "Private" level of protection?

- A. Data encapsulation B. Hiding data C. Limited manipulation ~~D. All A, B & C~~

Q-92: Which is the best way to provide data encapsulation?

- A. Make data protected B. Make data as package
C. ~~Make data private~~ D. Make data public

Q-93: Which of the following data separates design from implementation?

- A. Public ~~B. Private~~ C. Protected D. Package

8.10 Accessor and Mutator Methods

Q-94: Which method is used, when we want to allow others to use private data?

- A. ~~Accessor~~ B. Mutator C. Get D. Post

Q-95: Which method is used, when we want to allow others to modify private data?

- A. Accessor ~~B. Mutator~~ C. Private D. Public

Q-96: With which name accessor method is known due to convention?

- A. ~~Getter~~ B. Setter C. Package D. Public

Q-97: With which name mutator method is known due to convention?

- A. Post ~~B. Setter~~ C. Getter D. Private

Q-98: Which method is used if we want to allow other methods to read only the data value?

- A. ~~Getter~~ B. Setter C. Both A & B D. Package

Q-99: In Java, if we want to allow data directly accessed by methods of other class, then which method is used?

- A. Getter ~~B. Setter~~ C. Both A & B D. Package

Q-100: Which method is used if we want to allow other methods to modify the data value?

- A. Private B. Protected ~~C. Setter~~ D. Getter

Q-101: Which of the following method/s is/are used to prevent the variables from getting directly accessed and modified by other users of the class?

- A. Getter B. Setter ~~C. Both A & B~~ D. Package

8.11 Inheritance

Q-102: Which of the following allows us to build new class with added capabilities by extending existing class?

- A. Subclass ~~B. Inheritance~~ C. Relationship D. Method

Q-103: What is used to provide reusability feature of OOP language?

- A. Superclass B. Subclass C. Relationship ~~D. Inheritance~~

Q-104: Which kind of relationship is provided by inheritance models between two classes?

- A. Has ~~B. Is-a~~ C. Have D. Can

Q-105: Which symbol is used to show inheritance between two classes?

- A. Empty diamond B. Filled diamond
C. The arrow points from superclass to subclass ~~D. The arrow points from subclass to superclass~~

Q-106: Which class is referred to as superclass or base class?

- A. Inheritance B. First ~~C. Parent~~ D. Child

Q-107: Which class is referred to as subclass?

- A. First B. Main C. Parent ~~D. Child~~

Q-108: In which class common features are kept?

- ~~A. Superclass~~ B. Subclass C. Child class D. Common

Q-109: Which class inherits all instance variables and method from superclass and may have its own variables and methods?

- A. Child class B. Subclass ~~C. Both A & B~~ D. Superclass

Q-110: Which of the following are not inherited in subclass?

- A. Objects ~~B. Constructors~~ C. Variables D. Methods

Q-111: Which of the following keyword is used to create a subclass in class definition in Java?

- A. Void ~~B. Extends~~ C. Void main() D. Public static

Q-112: Which keyword is used to call the constructor of the superclass in the constructor of subclass?

- A. Extends B. Void C. Public static ~~D. Super~~

Q-113: Which of the following is not supported by Java?

- A. Extends keyword B. Multiple variables ~~C. Multiple inheritance~~ D. is-a Relationship

8.12 Composition and Aggregation

Q-114: Which of the following are the construction of classes that incorporate other objects?

- A. Composition B. Aggregation ~~C. Both A & B~~ D. Extends and Super

Q-115: Which kind of relationship is established between classes by composition and aggregation?

- A. Has-a B. Is-a C. Can D. Many